Channel-Level Unit Economics in Multi-Channel Restaurant Operations: A Profitability Analysis of In-Store, Aggregator, and Self-Delivery Sales at SkyCity Auckland

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***Abstract*—**Restaurants increasingly sell through four concurrent channels — in-store dining, Uber Eats, DoorDash, and self-managed delivery — each governed by a distinct cost structure. This paper analyzes 1,696 restaurant-branch monthly records from the SkyCity Auckland Restaurants & Bars dataset to quantify true channel-level profitability rather than order volume or revenue alone. We validate the dataset's net-profit formulas, normalize profit on a per-order basis, decompose costs into cost of goods sold (COGS), operating expense (OPEX), aggregator commission, and delivery-logistics components, and test sensitivity to commission-rate and delivery-cost changes. Aggregator channels (Uber Eats, DoorDash) are found to generate 61% of order volume and revenue but only 5% of net profit, at margins below 1%, because commission rates near 30% are layered on top of already high COGS and OPEX. In-Store and Self-Delivery channels, by contrast, produce approximately 95% of net profit from 39% of orders, at margins of 26.8% and 22.9% respectively. A commission-rate sensitivity model shows that a 5-percentage-point reduction would increase combined aggregator profit nearly sevenfold, while a 5-point increase would render aggregator channels loss-making. These findings support a shift in channel-prioritization strategy from volume-based to margin-based decision-making.

Index Terms*—channel profitability, unit economics, food delivery aggregators, commission drag, self-delivery ROI, restaurant analytics, margin sensitivity analysis.*

I. Introduction

A typical restaurant branch in this dataset serves customers who dine in, order through Uber Eats or DoorDash, or receive food from the restaurant's own delivery staff. Each channel carries a distinct cost structure: in-store dining has low variable costs but limited scalability; aggregators bring order volume but charge commissions reaching 30% of order value [1], [4]; self-delivery offers control over the customer relationship but introduces fixed per-order logistics costs. A restaurant chasing order growth without monitoring channel-level margins can scale a channel that is, transaction by transaction, losing money.

Prior operations-management literature has shown that the predominant industry contract — simple revenue sharing, under which the platform takes a fixed commission — fails to coordinate incentives between restaurants and platforms, systematically depressing restaurant margins on delivery orders [1]. Related work shows that adding a delivery channel materially reshapes a restaurant's optimal pricing and can be structurally unprofitable at high commission levels [2], while the choice between platform-provided and restaurant-operated ("self") logistics carries its own profitability and sustainability trade-offs [3]. This paper contributes an empirical, data-driven complement to that theoretical literature using a large real-world-style operational dataset.

A. Objectives

(1) Compare net profitability across In-Store, Uber Eats, DoorDash, and Self-Delivery channels; (2) quantify margin erosion from commissions and delivery logistics; (3) identify the most cost-efficient channel per restaurant, cuisine, and segment; (4) establish commission and delivery-cost sensitivity thresholds to support channel-prioritization and aggregator-negotiation decisions.

II. Dataset and Methodology

The dataset comprises 1,696 restaurant-branch monthly observations spanning 8 cuisine types, 4 business segments (Cafe, QSR, Full-service, Ghost Kitchen), and 4 Auckland subregions. For each branch, order count, revenue, and net profit are reported per channel, along with COGS rate (20–40%), OPEX rate (20–55%), aggregator commission rate, delivery radius, and self-delivery cost per order ($0.89–$5.31 NZD).

B. Profit Formula Validation

Each channel's net profit was recomputed from its stated cost-rate components and compared against the dataset's provided net-profit columns:

TABLE I. Profit Formula Validation

Channel	Formula	Max. diff.
In-Store	Rev×(1−C−O)	$0.005
Aggregators	Rev×(1−C−O−Comm)	$0.005
Self-Del.	Rev×(1−C−O)−SDcost	$0.005

(C = COGS rate, O = OPEX rate.) Discrepancies are consistent with floating-point rounding, confirming formula consistency. No missing values were found across the 30 columns and 1,696 rows.

C. Per-Order Normalization

Because channels differ in scale, profitability is normalized two ways: Net Profit per Order (channel profit ÷ channel orders) and Net Margin % (channel profit ÷ channel revenue). The former answers "what does one more transaction contribute"; the latter answers "what share of revenue survives as profit."

III. Channel Margin Analysis

Aggregating across all branches, the four channels show a wide profitability spread despite comparable revenue and order shares, as shown in Fig. 1.

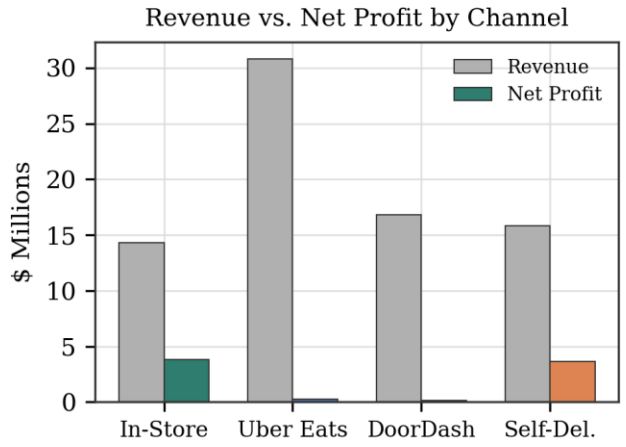


Fig. 1. Revenue vs. net profit by channel ($M). Aggregator revenue dwarfs profit.

TABLE II. Channel-Level Revenue, Profit, and Margin

Channel	Revenue	Profit	Margin
In-Store	$14.28M	$3.83M	26.80%
Uber Eats	$30.82M	$0.26M	0.84%
DoorDash	$16.79M	$0.15M	
Self-Del.	$15.85M	$3.63M	

Total portfolio revenue is $77.74M against $7.87M net profit — a blended margin of 10.1%. Fig. 2 makes the margin gap explicit: aggregator margins sit below 1% while In-Store and Self-Delivery exceed 20%.

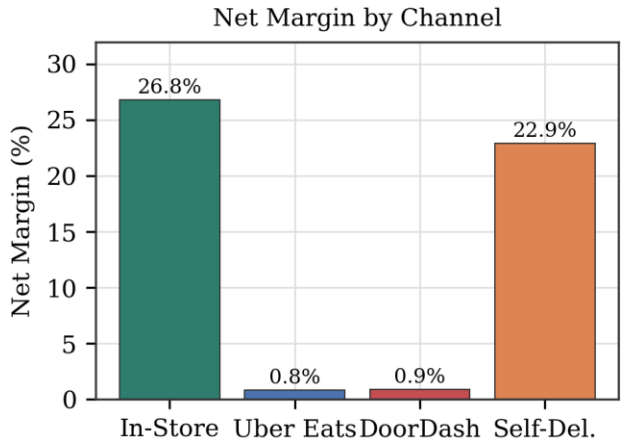


Fig. 2. Net margin (%) by channel. Aggregator margins are compressed to near break-even.

The gap between revenue share and profit share is the central finding of this study:

TABLE III. Share of Orders, Revenue, and Profit

Channel	Orders	Revenue	Profit
In-Store	18.4%	18.4%	48.7%
Uber Eats	39.6%	39.6%	
DoorDash	21.6%	21.6%	
Self-Del.	20.4%	20.4%	4

Combined, the two aggregators generate 61.2% of orders and revenue while contributing 5.2% of profit. In-Store and Self-Delivery, at 38.8% of orders, produce 94.8% of profit — direct empirical evidence that order volume is a poor proxy for channel value, consistent with the coordination failure predicted by revenue-sharing contract theory [1].

IV. Cost Component Decomposition

Average commission on aggregator orders is approximately 30% of revenue (range 27–33%), layered on top of COGS (24–38%) and OPEX (27–50%). Total commissions paid across the portfolio were $14.29M ($9.25M to Uber Eats, $5.04M to DoorDash) — 1.8× the entire portfolio's net profit. Self-delivery logistics cost totaled $1.28M against $15.85M revenue (≈8.1% effective rate), far below the aggregator commission rate, explaining why self-delivery margin (22.9%) tracks closely with in-store margin (26.8%) rather than with aggregator margin (≈0.9%).

V. Sensitivity Analysis

Combined Uber Eats and DoorDash profit was simulated under uniform commission-rate shifts, shown in Fig. 3.

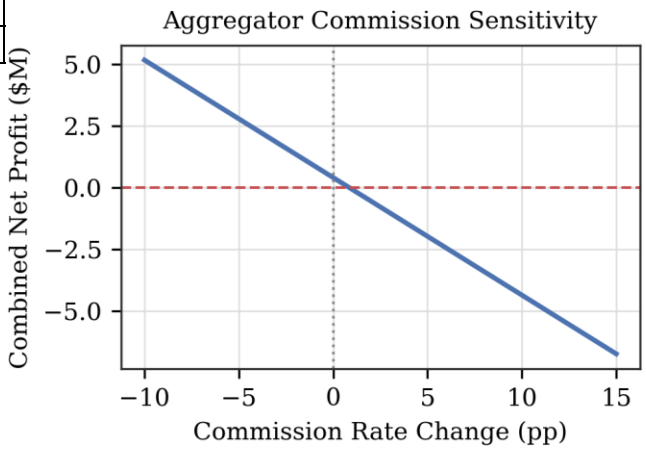


Fig. 3. Combined aggregator net profit as commission rate shifts. The channel turns loss-making beyond roughly +3pp.

TABLE IV. Commission Sensitivity (Combined Aggregators)

Δ Commission	Combined net profit
−5 pp	$2.79M (+581%)
0 (baseline)	$0.41M
+5 pp	−$1.97M
+10 pp	−$4.35M

A 5-point commission increase (e.g., 30%→35%) converts combined aggregator channels from marginally profitable to a $2M loss; a 5-point reduction increases profit nearly sevenfold. For self-delivery, modeling profit-per-order against rising delivery cost (holding average revenue-per-order at $38.54 and combined COGS+OPEX at ≈69.2% constant) yields a breakeven delivery cost

of approximately $11.88/order — versus an observed average of $3.12/order, a 3.8× margin of safety.

VI. Cuisine, Segment, and Regional Effects

Blended margin varies by cuisine from 5.3% (Kebabs/Mediterranean) to 14.5% (Pizza), with Burgers (12.4%) also above the portfolio mean. Segment-level variation is more pronounced, as Fig. 4 illustrates:

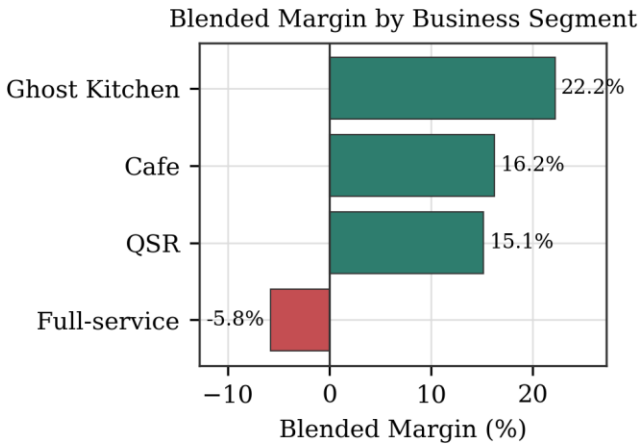


Fig. 4. Blended margin by business segment. Full-service is the only segment operating at a portfolio-wide loss.

TABLE V. Blended Margin by Segment

Segment	Margin	COGS	OPEX
Full-service	−5.8%	37.5%	50.0%
QSR	15.1%	23.9%	40.2%
Cafe	16.2%	24.1%	39.8%
Ghost Kitchen	22.2%	27.6%	27.1%

Full-service is the only segment with a negative blended margin, driven by structurally higher COGS and OPEX rates relative to other business models. Regional variation across Auckland subregions is comparatively modest (9.5%–10.5%), indicating channel mix and segment composition are stronger profitability drivers than geography in this dataset.

VII. Profit Volatility and Risk

Margin stability was assessed via coefficient of variation (CoV = σ/μ) of channel margin % across restaurants. Fig. 5 shows the full margin distribution per channel, including spread and outliers.

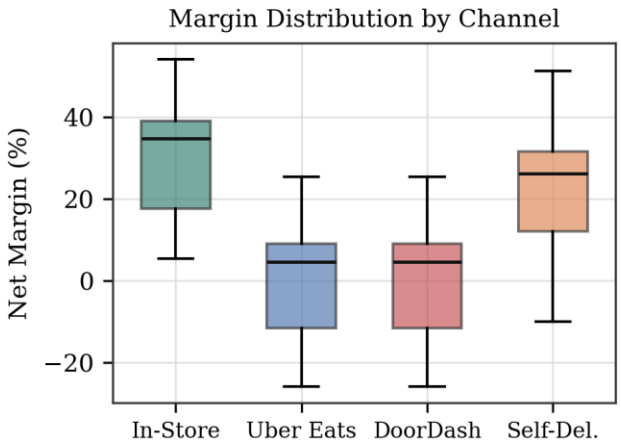


Fig. 5. Distribution of net margin (%) by channel. Aggregator margins straddle zero with wide dispersion.

TABLE VI. Margin Volatility and Loss Rates

Channel	Mean	CoV	% Loss-making
In-Store	30.8%	0.39	0.0%
Self-Del.	22.6%	0.56	5.3%
Uber Eats	0.8%	15.23	31.7%
DoorDash	0.8%	15.23	31.7%

Aggregator-channel volatility scores are roughly 27–39× higher than In-Store, because near-zero mean margin makes branches highly sensitive to small cost-rate variation. Nearly one-third of branches already operate at a net loss on aggregator orders before any adverse commission change, consistent with prior findings that unmodified revenue-sharing contracts leave restaurant margins acutely exposed [1], [2].

VIII. Discussion and Limitations

These results represent a cross-sectional snapshot of monthly unit economics, not a causal study. The dataset does not capture customer-acquisition value from aggregator exposure, brand-visibility effects, or fixed costs of building an in-house delivery fleet beyond the modeled per-order cost. Restaurants should weigh these strategic factors alongside the unit-economics evidence presented here.

IX. Recommendations

(1) Prioritize In-Store and Self-Delivery growth, which already generate 94.8% of profit from 38.8% of orders; (2) treat aggregator growth as volume rather than profit growth absent commission concessions; (3) use the quantified sensitivity curve as a negotiation anchor — a 5-point concession is worth ≈$2.4M in annualized profit; (4) expand self-delivery capacity, given 3.8× headroom to the estimated breakeven cost; (5) target Full-service segment economics for cost-structure review, given its portfolio-wide negative margin; (6) flag the ≈32% of branches already loss-making on aggregator channels for pricing or channel-mix intervention.

X. Conclusion

Across 1,696 restaurant-branch records, order volume and channel profitability are shown to be only weakly related. Aggregator channels generate the majority of orders and revenue but a small fraction of profit, consistent with theoretical predictions that simple

commission-based revenue sharing does not protect restaurant margins [1]. In-Store dining and self-managed delivery are the portfolio's true profit engines. The evidence supports a strategic shift toward margin-aware channel management, with concrete, quantified thresholds to guide commission negotiation and self-delivery investment.

Acknowledgment

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